

Financial Risk Analysis with Python

Customer Financial Behavior & Risk Analysis Report

Prepared for: Morgan Stanley — Executive Leadership Team

Role: Financial Data Analyst

Tools Used: Python (Pandas, NumPy, Regex, Matplotlib/Seaborn, SciPy)

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Project Overview

You are hired as a Financial Data Analyst by Morgan Stanley, a major global banking institution. As digital transactions rise and customers diversify their account usage, the executive team wants to uncover patterns in customer financial behavior, account performance, and transaction risks in order to improve customer service and minimize financial risk. Leadership is particularly interested in the following questions:

- How do customers interact with different account types (e.g., savings, current, credit)?
- What are the trends in debit vs. credit transactions across segments?
- Can accounts with unusually high risk or inconsistent financial activity be identified?
- What transaction behaviors correlate with lower account balances or overdraft incidents?

Objective

Build a complete Customer Financial Behavior and Risk Analysis Report using Python for data handling, consolidating the findings into a single PDF report. The report cleans and consolidates Morgan Stanley's customer account data, uncovers transaction insights, and helps decision-makers align customer strategies with usage trends and financial risk indicators.

Dataset

Morgan Stanley Financial Transactional Data. A transactional summary of customer accounts, including account numbers, transaction dates, transaction types (credit/debit), transaction amounts, account types, and available balances.

Task 1 — Data Cleaning and Formatting

OBJECTIVE

Improve the quality of the raw dataset before performing any analysis.

WORK PERFORMED

- Imported required libraries such as Pandas, NumPy, and Regular Expressions.
- Loaded the dataset into a DataFrame.
- Inspected data types and descriptive statistics.
- Removed special characters from financial columns.
- Converted financial fields into numeric format.
- Validated transaction dates.
- Converted dates into standard YYYY-MM-DD format.
- Checked unique Account Types and Transaction Types for consistency.
- Saved the cleaned dataset for the next stage of analysis.

KEY INSIGHT

The cleaned dataset became consistent, reliable, and suitable for further analysis.

Task 2 — Descriptive Transaction Analysis

OBJECTIVE

Analyze customer transaction patterns over time and identify account behavior.

WORK PERFORMED

- Created Year and Month columns.
- Calculated monthly credit transactions.
- Calculated monthly debit transactions.
- Calculated monthly net transaction volume.
- Generated a monthly Credit vs. Debit trend chart.
- Calculated net inflow for every customer account.
- Identified top-performing accounts.
- Identified bottom-performing accounts.
- Calculated transaction gaps.
- Flagged dormant accounts when the gap between transactions was approximately 60 days or more.

KEY INSIGHTS

- *Monthly transaction summaries help identify seasonal transaction patterns.*
- *Credit and debit trend analysis helps understand customer spending behavior.*
- *Net inflow identifies profitable and low-performing accounts.*
- *Dormant account detection helps identify inactive customers for future engagement campaigns.*

Task 3 — Customer Profile Building

OBJECTIVE

Categorize customers based on transaction behavior and account balance.

ACTIVITY LEVEL CLASSIFICATION

Activity Level	Rubric (Transaction Frequency)
High Activity	6 or more transactions
Medium Activity	3 – 5 transactions
Low Activity	Fewer than 3 transactions

CUSTOMER SEGMENTATION

Customers were segmented based on **Average Account Balance** and **Transaction Volume** into three resulting segments:

- Premium
- Regular
- Basic

CUSTOMER PROFILING

- High Net Inflow Accounts
- High Frequency Low Balance Accounts
- Negative Balance Accounts
- Near-Zero Balance Accounts

KEY INSIGHTS

- *Customers with higher balances and frequent transactions were categorized as Premium customers.*
- *Customers with high transaction frequency but lower balances may require closer financial monitoring.*
- *Accounts with negative or near-zero balances indicate potential financial risk.*

Task 4 — Financial Risk Identification

OBJECTIVE

Identify customers or accounts that may pose a financial risk based on their transaction behavior.

WORK PERFORMED

- Detected accounts with large withdrawals (■75,000 or more).
- Flagged accounts making frequent large withdrawals (2 or more).
- Identified overdraft accounts (negative account balance).
- Calculated balance volatility using Standard Deviation and Coefficient of Variation.
- Detected anomalies (outliers) using the IQR (Interquartile Range) method and Z-Score.
- Combined all risk indicators to create a SuspiciousStatus column.
- Marked risky customers as “Suspicious” and others as “Normal”.

KEY INSIGHT

Customers with repeated large withdrawals, overdrafts, or abnormal transaction amounts are more likely to require monitoring for potential financial risk.

Task 5 — Exploratory Data Analysis (EDA) & Visualization

OBJECTIVE

Understand customer transaction behavior using charts and visual analysis.

WORK PERFORMED

- Reviewed dataset structure, summary statistics, and missing values.
- Visualized Transaction Type Distribution.
- Visualized Account Type Distribution.
- Visualized Top 10 and Bottom 10 Net Inflow Accounts.
- Visualized Customer Activity Levels and Customer Segments.
- Visualized Account Balance Distribution and Transaction Amount Distribution.
- Visualized Suspicious vs. Normal Customers.
- Generated a Correlation Heatmap.
- Plotted Transaction Amount Outliers (Boxplot).
- Plotted Balance vs. Transaction Amount (Scatter Plot).
- Compared Dormant vs. Active Accounts.

KEY INSIGHT

The visualizations helped identify customer behavior patterns, active and inactive accounts, high-performing customers, suspicious transactions, and relationships between financial variables.

Task 6 — Hypothesis Testing

OBJECTIVE

Use statistical tests to determine whether observed differences are statistically significant.

TEST 1 — HIGH VS. LOW TRANSACTION VOLUME

- Divided accounts into High-volume accounts (6 or more transactions) and Low-volume accounts (2 or fewer transactions).
- Used an Independent t-test to compare their average balances.
- Checked the p-value to decide whether to reject or accept the null hypothesis.

TEST 2 — CUSTOMER SEGMENTS

- Compared Premium, Regular, and Basic customers.
- Used One-Way ANOVA to determine whether average balances differ among the three groups.
- Interpreted the result using the p-value.

KEY INSIGHT

Hypothesis testing determines whether differences in average balances are statistically significant or simply due to random variation.

Conclusion

This project successfully transformed raw financial transaction data into meaningful business insights using Python. Data preprocessing, descriptive analysis, and customer profiling provided valuable information about customer behavior, account performance, and potential financial risks. These insights can assist financial institutions in improving customer engagement, monitoring risky accounts, and supporting data-driven decision-making.